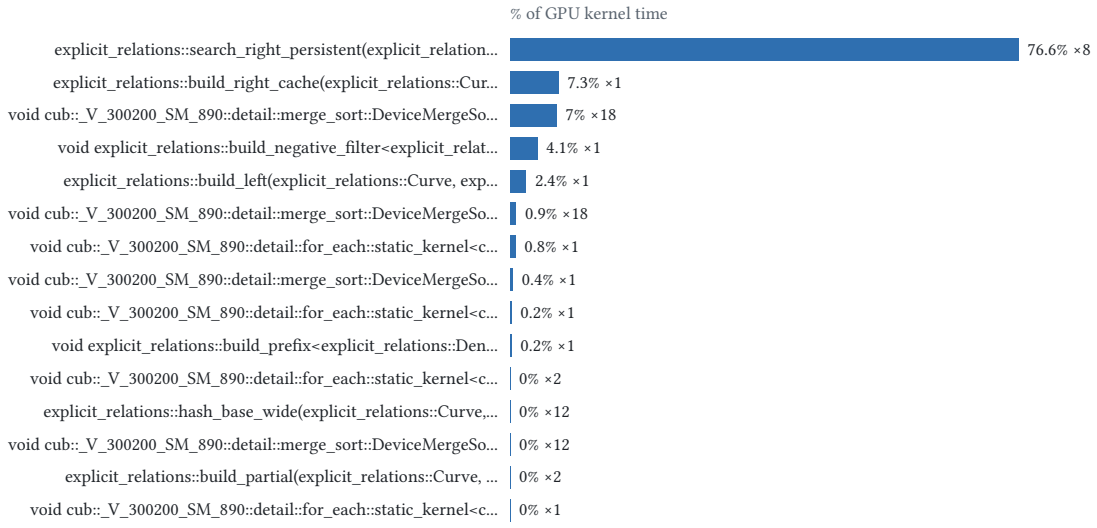
GPU query timeline follow-up

RTX 4090 · fixed 63-bit query prefixes · 31 October 2026

Both Nsight Systems traces completed. Their service replies, and both separate unprofiled replays, matched the saved v1 prefixes byte for byte. Each prefix contains eight full-table misses and includes one-time table preparation.

Native trace observation	Seed 101	Seed 202
Kernel coverage of first-to-last kernel span	99.10%	98.70%
Search-kernel launches	8	8
Search share of summed kernel time	76.6%	50.5%
Search total (profiled milliseconds)	1338.874	695.057

GPU time by kernel (nsys)



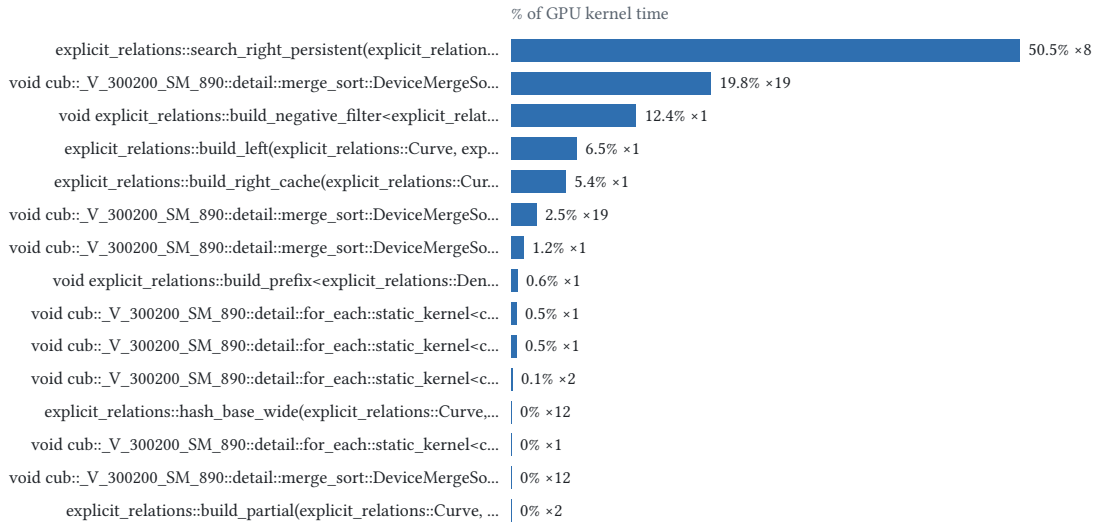
nsys cuda_gpu_kern_sum; run nsys-101

Seed 101. The original persistent search kernel is the largest trace row. The GPU had a kernel active for 99.1% of the span between its first kernel start and last kernel end. This is kernel coverage, not measured SM utilization. Launch gaps are small within this trace. One-time table construction is included, so its setup/query shares must not be transferred to a full recovery run.

Copies. The host cudaMemcpy API total includes waiting for the search result. The GPU memory trace for seed101 records only 0.033888 ms device-to-host and 0.000736 ms host-to-device; API wait time is not bulk-transfer cost.

Second input and counter availability

GPU time by kernel (nsys)



nsys cuda_gpu_kern_sum; run nsys-202

Seed 202. Kernel coverage was 98.7%. Table preparation occupies a larger fraction of this eight-query trace than for seed101. These profiler durations are diagnostic values, not production throughput or a cross-host comparison.

Hardware counters. Both Nsight Compute attempts returned unsupported with ERR_NVGPUCTRPERM. The separate probe built, ran and checked its control kernel, but encountered the same provider counter restriction. No hardware instruction, occupancy or warp-stall counters were collected or estimated. No driver setting was changed. The block-clock results from v1 remain separately preserved.

Accounting. Six planned process attempts: two unprofiled exact-reply successes, two successful timelines, two unsupported counter attempts. The capability probe is separate. Native summaries, SVGs and report sources regenerated byte-identically. Canonical export/schema/coverage/file checks passed with zero findings. The owned pod fmu20lo21ib6gu was deleted and absence verified; the v1 pod was already gone.

Reproduction. CUDA 13.2 pinned image; Nsight Systems 2025.6.3 and Compute 2026.1.1, installed via pinned 13.2.2 metapackages. The launch manifest hashes bash; the actual GPU service hash is in instruments.sha256. Source b50e30d47458881eba10d802b7f972a9f56541bd. Protocol SHA256: 4dc4c0560a8af2ad967d99c9f1d08526eb676e435b12ef658a174e093eba869e. Study: research/explicit_relations_profile_20261031; raw results-v2/export.json.

Next useful construction. Isolate an uninstrumented full-cache specialization and measure it on matched complete recoveries. The instrumented query breakdown puts lookup and normalization ahead of coordinate conversion; hardware stall attribution remains unavailable on this provider configuration.